

— SUPPLEMENTARY INFORMATION —

**LOCAL CERTIFICATES FOR COMPENSATED GRAM BOUNDS:  
THE TWO SEVEN-POINT INPUTS AND THE COMPLETED ONE-SHORT  
THIRTEEN-POINT REGIONS**

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ABSTRACT. This supplement gives the mathematical specification and acceptance proofs for two computer-assisted inequalities on six nonnegative gaps. They are the cosine row with floor  $19/5000$  and pressure  $1/3000$ , and the two-mode row with floor  $7/400$  and pressure  $1/500$ . We explain interval arithmetic, removable singularities, whole-cell range minima, matrix lower bounds, and complete-domain accounting, with source-bound records of completed executions. A separate six-mode, thirteen-point problem is developed from its scalar exclusions and two analytic baselines. An adjacent-long-gap certificate and four completed triple bounds then exclude its all-long region and every one-short region, including both interior placements. The remaining multi-short region is not certified. No full thirteen-point inequality or additional global zero-count bound is asserted.

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## 1. SCOPE, DEPENDENCIES, AND THE MEANING OF A CERTIFICATE

There are three different Fourier kernels in this supplement. The first two supply the only local numerical inputs to the main manuscript. The third concerns a separate, unfinished thirteen-point problem. A floor, pressure allowance, or quadratic reserve proved for one kernel is not available for either of the others.

All the mathematics needed to understand the local certificates is given below. The accompanying verification package is a computational implementation of these arguments, not a substitute for them. Its manifest identifies the distributed sources and the provenance of recorded results. A digest establishes byte identity; by itself it does not establish an interval enclosure or a covering argument.

The trust boundary has three parts. First are the displayed analytic and finite-dimensional arguments. Second are a correctly implemented outward arithmetic backend and the source implementing the stated acceptance and coverage rules. Third is an actual completed execution of that source with the exact configuration in question. The recorded executions used Arb through `python-flint` version 0.9.0 at 128-bit precision for scalar cell tables; the later triple certificates use exact rational arithmetic after these tables have been established. None of these computations is presented as a formal proof-assistant verification.

The scalar-cell, range-minimum, and convex-tangent verification framework, including the cosine seven-point row, comes from Ainta's MIT-licensed implementation [1]. The two-mode callbacks and the additional acceptance rules are adaptations. The Fourier-profile and local-pressure viewpoint is not claimed here as an invention of this supplement. The main manuscript discusses its mathematical prior art and the distinct common-witness transfer. Here the purpose is to expose the precise local proof obligations and their evidence, including the limits of that evidence.

## 2. THE KERNELS AND THE LOCAL ENERGY

Put  $z = 1/\sqrt{2}$  and define the normalized cosine density by

$$(2.1) \quad f_0(t) = \frac{\cos(\sqrt{2}t)}{\sqrt{2} \sin z} \quad (|t| \leq \tfrac{1}{2}), \quad f_0(t) = 0 \quad (|t| > \tfrac{1}{2}).$$

For a finite real vector  $u = (u_1, \dots, u_d)$ , set

$$(2.2) \quad f_u(t) = f_0(t) + \sum_{j=1}^d u_j \cos(2\pi j t) \quad (|t| \leq \tfrac{1}{2}),$$

and set it to zero elsewhere. Direct integration gives  $\int f_0 = 1$ ; each perturbing cosine has integral zero. Thus  $\int f_u = 1$ . The three choices used here are

$$(2.3) \quad u_C = (), \quad u_H = \left( \frac{13}{500}, -\frac{7}{500} \right), \quad u_{13} = \frac{1}{4000}(37, -69, -1, 8, -11, 2).$$

The elementary inequalities  $\cos z \geq 3/4$  and  $0 < \sqrt{2} \sin z \leq 1$  imply  $f_0 \geq 3/4$  on its support. The absolute coefficient sums in the last two profiles are respectively  $1/25$  and  $4/125$ . Consequently all three densities are nonnegative; in particular  $f_{u_{13}} \geq 359/500 > 0$ . Each is real and even.

Our Fourier convention and squared kernel are

$$(2.4) \quad k_u(x) = \int_{-1/2}^{1/2} f_u(t) e^{-2\pi i x t} dt, \quad q_u(x) = k_u(x)^2 \quad (x \in \mathbb{R}).$$

The kernel is real on the real axis, and  $q_u \geq 0$ . Writing  $\text{sinc } y = \sin y / y$  with  $\text{sinc } 0 = 1$ , direct integration yields the entire formulas

$$(2.5) \quad k_0(x) = \frac{\text{sinc}(\pi x - z) + \text{sinc}(\pi x + z)}{2\sqrt{2} \sin z},$$

$$(2.6) \quad k_u(x) = k_0(x) + \frac{1}{2} \sum_{j=1}^d u_j \{ \text{sinc}(\pi(x - j)) + \text{sinc}(\pi(x + j)) \}.$$

These formulas have no actual poles, even when a shifted argument vanishes. Since the support is compact, differentiation under the integral is valid to every order. Positivity and normalization give

$$(2.7) \quad |k_u| \leq 1, \quad |k'_u| \leq \pi, \quad |k''_u| \leq \pi^2, \quad |q''_u| \leq 4\pi^2.$$

The last inequality follows from  $q''_u = 2(k'_u)^2 + 2k_u k''_u$ .

For  $m - 1$  nonnegative gaps  $g = (g_1, \dots, g_{m-1})$ , define

$$(2.8) \quad V_{m,u}(g) = \sum_{h=1}^{m-1} \frac{2}{m-h} \sum_{i=1}^{m-h} q_u(g_i + \dots + g_{i+h-1}).$$

Zero gaps are included. The coefficients in this formula are part of the objective and must not be replaced by uniform coefficients over all pairs. The two seven-point assertions are

$$(2.9) \quad V_{7,C}(g) + \frac{1}{3000} \sum_{j=1}^6 g_j \geq \frac{19}{5000},$$

$$(2.10) \quad V_{7,H}(g) + \frac{1}{500} \sum_{j=1}^6 g_j \geq \frac{7}{400},$$

for every  $g \in [0, \infty)^6$ . These are whole-orthant statements, not inequalities only at a mesh or at distinct configurations.

### 3. ARITHMETIC THAT ENCLOSES FUNCTIONS, NOT SAMPLES

**3.1. Outward operations and removable singularities.** An interval  $[a, b]$  denotes a set containing the exact quantity in question. Sums, differences, products, and quotients are rounded outwards; division is allowed only after zero has been excluded from the denominator interval. For rational intervals the endpoints and all comparisons can be exact. For Arb balls the containment guarantee of the backend is used, followed, where required, by a downward conversion to the exact real number represented by a binary64 value. Converting a lower endpoint to nearest binary64 and then applying `nextafter` toward  $-\infty$  gives a conservative lower endpoint when the result is finite. A nonnegative lower bound is clipped at zero only when nonnegativity has independently been proved.

Away from zero, the identities

$$\begin{aligned} \text{sinc}'(y) &= \frac{y \cos y - \sin y}{y^2}, \\ \text{sinc}''(y) &= \frac{(2 - y^2) \sin y - 2y \cos y}{y^3} \end{aligned}$$

may be evaluated by interval arithmetic. If an argument ball includes zero, these divisions are not permitted. The following explicit polynomials handle such balls. For  $|y| \leq b < 1$ , let

$$\begin{aligned} P_0(y) &= \sum_{n=0}^8 \frac{(-1)^n y^{2n}}{(2n+1)!}, \\ P_1(y) &= - \sum_{n=0}^8 \frac{(-1)^n y^{2n+1}}{(2n+1)!(2n+3)}, \\ P_2(y) &= - \sum_{n=0}^8 \frac{(-1)^n y^{2n}}{(2n)!(2n+3)}. \end{aligned}$$

Then the respective error bounds are

$$(3.1) \quad |\text{sinc} - P_0| \leq \frac{b^{18}}{19!}, \quad |\text{sinc}' - P_1| \leq \frac{b^{19}}{19! 21}, \quad |\text{sinc}'' - P_2| \leq \frac{b^{18}}{18! 21}.$$

For completeness,  $\text{sinc } y = \int_0^1 \cos(ty) dt$ . Differentiate this identity once or twice and apply the real Taylor remainder for sine or cosine inside the integral. Integrating the extra powers of  $t$  gives exactly the three denominators in (3.1). Thus these are bounds over the whole ball, not an alternating-series test at its center. Substitution in (2.5)–(2.6), with factors  $\pi$  and  $\pi^2$  for differentiation in  $x$ , encloses  $k, k', k''$  and hence  $q, q', q''$  for each of the three specified profiles.

**3.2. A completely rational point evaluator.** The scalar gates and the atomic Taylor certificates below can be checked without floating transcendental arithmetic at their rational evaluation points. Here is an explicit construction. The Machin identity

$$(3.2) \quad \pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and the sums  $\sum_{n=0}^{40} (-1)^n / ((2n+1)a^{2n+1})$  give rational enclosures for the two arctangents. Each error is at most  $1/(83a^{83})$ , with its alternating sign. Combine their endpoint intervals outwards. If desired, after each operation round to the outward rational lattice of spacing  $10^{-50}$ .

Put

$$(3.3) \quad R_0 = \frac{1}{2} + z \cot z.$$

The Taylor sums through index 30 for  $\cos z$  and  $\sin z/z$ , with the next alternating term, enclose them using only powers of  $z^2 = 1/2$ . Divide these positive rational intervals to enclose  $z \cot z$ . No numerical square root is needed for this operation.

For rational  $x$ , write  $x = n + v$ , where  $n = \lfloor x + 1/2 \rfloor$  and  $-1/2 \leq v < 1/2$ . Evaluate  $\sin(\pi v)$  through degree 47 and  $\cos(\pi v)$  through degree 46, including symmetric remainders  $2^{49}/49!$  and  $2^{48}/48!$ . These follow from  $|\pi v| < 2$  and the ordinary real Taylor theorem. Interval evaluation of these polynomials, and the factor  $(-1)^n$ , encloses  $\sin(\pi x)$  and  $\cos(\pi x)$ .

The directly integrated point formulas are

$$(3.4) \quad k_0(x) = \frac{\cos(\pi x) - 2(R_0 - 1/2)\pi x \sin(\pi x)}{1 - 2\pi^2 x^2},$$

$$(3.5) \quad k_u(x) - k_0(x) = \sum_{j=1}^d u_j \frac{(-1)^j x \sin(\pi x)}{\pi(x^2 - j^2)}.$$

At  $x = j$  the corresponding mode integral is exactly  $1/2$  and is handled separately. Other denominators must be checked to exclude zero. All rational point evaluations below have  $x \geq 5/6$ . The atomic helper includes the separately guarded range  $[9/10, 1)$ . On that below-one range  $1 - 2\pi^2 x^2 < 0$  and each  $x^2 - j^2 < 0$ ; the interval implementation still verifies these signs before division. At the positive integers the removable resonances are handled explicitly. This is a point

evaluator with proved error bounds. A whole interval is enclosed either by the entire sinc formulas or by an additional derivative bound; point evaluation alone is never a domain certificate.

#### 4. THE COMPLETE SEVEN-POINT VERIFICATION

**4.1. Pressure compactification and scalar cell tables.** Write  $F_7 = V_7 + p \sum g_j$ , with one of the exact pairs  $(\varepsilon, p)$  in (2.9)–(2.10). Since  $V_7 \geq 0$ , the region  $\sum g_j \geq \varepsilon/p$  is settled by pressure alone. Every strict counterexample would therefore have all its coordinates less than  $\varepsilon/p$ . The common grid is  $G = 4000$ , with closed cells

$$(4.1) \quad J_i = [i/G, (i+1)/G] \quad (i = 0, 1, \dots).$$

Let  $L = \lceil \varepsilon G/p \rceil$ . The coordinate cells  $0, \dots, L-1$  cover every possible strict counterexample; shared boundaries are included. The recorded tables have  $L+8$  entries. They supply lower bounds

$$(4.2) \quad q(x) \geq Q_i \geq 0, \quad q''(x) \geq D_i \quad (x \in J_i)$$

where the second assertion is made only for finite, available entries. To obtain  $Q_i$ , enclose  $k(J_i)$  by Arb, take a nonnegative lower bound for its absolute value, convert downwards, and square downwards. To obtain  $D_i$ , enclose  $q''(J_i)$  and convert its lower endpoint downwards. An unavailable curvature prefix is represented by  $-\infty$ , not zero.

It is unnecessary for these tables to contain every possible summed distance in the six-dimensional box. If a sum reaches beyond the stored  $q$  table, the universally valid lower bound zero is used for that energy term. If it reaches beyond the stored curvature table, the corresponding derivative test is unavailable. No extrapolation or unproved zero-curvature assertion is made.

Before forming multidimensional roots, remove any coordinate cell for which

$$(4.3) \quad \frac{1}{3}Q_i + p\frac{i}{G} \geq \varepsilon.$$

This is valid because that coordinate's nearest-gap term in  $V_7$  has coefficient  $1/3$ , and every discarded term of  $F_7$  is nonnegative. Maximal consecutive surviving groups form scalar components. Their Cartesian products cover all remaining candidates.

**4.2. Range indices and elementary lower bounds.** Represent a box by six inclusive integer ranges  $[a_j, b_j]$ . Its physical coordinate interval is  $[a_j/G, (b_j+1)/G]$ . A sum of  $h$  consecutive coordinates is contained in the scalar cells from

$$(4.4) \quad a_i + \dots + a_{i+h-1} \quad \text{through} \quad b_i + \dots + b_{i+h-1} + h - 1,$$

inclusive. The final  $h-1$  is essential. Omitting it would leave part of the closed sum interval unchecked.

Taking the minimum of every  $Q$  entry in this range gives a whole-box lower bound for its energy term. A sparse range-minimum structure can answer the same query using two overlapping power-of-two ranges whose union is the entire requested range. Overlap is harmless for a minimum; a gap would not be. Adding all the resulting lower bounds with the original positive coefficients and the pressure at lower coordinate endpoints gives a valid interval lower bound for  $F_7$ . The target is rounded upwards and lower contributions downwards before any binary64 acceptance comparison.

**4.3. Signed Hessian bounds and center tests.** For each pair distance write  $x_e = v_e^T g$ , where  $v_e$  is the zero-one incidence vector of its consecutive gaps, and let  $c_e = 2/(7-h_e)$ , where  $h_e = v_e^T v_e$ . The exact Hessian is

$$(4.5) \quad \nabla^2 F_7(g) = \sum_e c_e q''(v_e^T g) v_e v_e^T.$$

If  $\ell_e \leq c_e q''(v_e^T g)$  throughout a box, then the exact represented matrix

$$(4.6) \quad H_- := \sum_e \ell_e v_e v_e^T \preceq \nabla^2 F_7(g)$$

throughout it. Indeed, the difference is a sum of positive semidefinite rank-one matrices with nonnegative coefficients. Negative  $\ell_e$  are allowed. When  $c_e$  is bracketed by binary64 endpoints, its lower endpoint must be used for a nonnegative curvature lower bound and its upper endpoint for a negative one; the product is then rounded downwards. This sign rule prevents an upward error in a negative Hessian contribution.

Let  $c$  be the rational center of the physical box, and let  $r_j$  be its rational coordinate radii. If rigorous LDL decomposition establishes  $H_-$  positive definite,  $F_7$  is convex on the box. In particular,

$$(4.7) \quad F_7(g) \geq F_7(c) - \sum_{j=1}^6 |\partial_j F_7(c)| r_j.$$

The center value is bounded below and each derivative magnitude above by Arb. A floating LDL test may be used to decide whether to attempt the rigorous test, but it cannot accept a box. The rigorous LDL test uses the exact represented rational matrix and requires every pivot strictly positive, with all interval divisions justified.

The two-mode computation also uses the stronger lower quadratic. Taylor's integral remainder along the segment from  $c$  to  $g$  gives

$$(4.8) \quad F_7(g) \geq F_7(c) + \nabla F_7(c)^T (g - c) + \frac{1}{2} (g - c)^T H_- (g - c).$$

If  $H_- = LDL^T$  is positive definite, minimizing the right-hand side over all of  $\mathbb{R}^6$  gives

$$(4.9) \quad F_7(g) \geq F_7(c) - \frac{1}{2} \nabla F_7(c)^T H_-^{-1} \nabla F_7(c).$$

Solving  $Ly = \nabla F_7(c)$  by interval forward substitution encloses the quadratic term as  $\sum_j y_j^2 / D_j$ . It is subtracted with an upper bound. The minimizing point need not belong to the box.

A further valid fallback discards positive curvature contributions. For  $\ell_e < 0$ , Cauchy-Schwarz gives  $\ell_e v_e v_e^T \succeq \ell_e h_e I$ . Thus  $\mu I \preceq H_-$  with  $\mu = \sum_{\ell_e < 0} h_e \ell_e \leq 0$ . The lower Taylor bound becomes

$$(4.10) \quad F_7(g) \geq F_7(c) - \sum_j |\partial_j F_7(c)| r_j + \frac{\mu}{2} \sum_j r_j^2.$$

All quantities are enclosed outwards. Each of (4.7), (4.9), and (4.10) is an alternative lower bound. Their maximum is permissible; adding them would count the same cost more than once.

Center values may be cached because every center distance is exactly of the form  $k/(2G)$  for an integer  $k$ . A cache key must encode this exact index, and the profile and precision must be immutable during the run. Caching complete enclosing balls does not change any acceptance rule.

**4.4. Exhaustion of the domain.** If a box passes none of the valid lower bounds, bisect its widest integer coordinate range. For  $a < b$ , the two children  $[a, m]$  and  $[m + 1, b]$ ,  $m = \lfloor (a + b)/2 \rfloor$ , partition its atomic cell indices and cover its closed physical interval. Continue until all roots have been discharged. A remaining atomic box that passes no test is unresolved, not accepted and not automatically a counterexample. Similarly, a node cap, exception, or interrupted execution is not a successful proof.

Uniform pressure and (2.8) are invariant under full reversal of the six gaps. A reversal reduction, if used, must keep one representative for each genuinely reversed pair. There is no arbitrary permutation symmetry. Source review and complete tree traversal establish coverage; identities such as

$$(4.11) \quad \text{roots} + \text{splits} = \text{accepted leaves}, \quad \text{visited nodes} = \text{splits} + \text{accepted leaves}$$

are useful independent consistency checks on a completed binary forest, but are not a replacement for that coverage argument.

## 5. COMPLETED VERIFICATION OF THE TWO SEVEN-POINT ROWS

The distributed verification package recomputed both seven-point inequalities with the exact configurations above. The following table records these completed executions. Both use grid 4000, 128-bit Arb arithmetic, and curvature start index 1001. Each successful execution exhausted its covering forest with no unresolved atomic cell.

| Quantity               | Cosine row | Two-mode row |
|------------------------|------------|--------------|
| Exact floor            | 19/5000    | 7/400        |
| Pressure per gap       | 1/3000     | 1/500        |
| Grid / Arb precision   | 4000/128   | 4000/128     |
| Coordinate cutoff $L$  | 45600      | 35000        |
| Scalar table entries   | 45608      | 35008        |
| Curvature start index  | 1001       | 1001         |
| Initial roots          | 378        | 1            |
| Visited nodes          | 268856     | 129093       |
| Splits                 | 134239     | 64546        |
| Accepted leaves        | 134617     | 64547        |
| Pressure leaves        | 1597       | 602          |
| Interval leaves        | 111698     | 38533        |
| Derivative-test leaves | 21322      | 25412        |
| Maximum depth          | 32         | 59           |
| Unresolved / pending   | 0/0        | 0/0          |

The cosine coordinate components, in inclusive cell indices, are

$$(5.1) \quad [3809, 4778], \quad [7221, 9363], \quad [10572, 44827].$$

There are  $3^6 = 729$  Cartesian roots, of which  $(729 + 27)/2 = 378$  represent the reversal orbits. The two-mode computation has the single component  $[3443, 34516]$  and therefore one six-dimensional root. Its category named “tangent” in the saved report includes the valid derivative alternatives above; that label is not a claim that each such leaf used only the affine tangent formula. Its center-cache report has 1666485 hits and 2574 misses.

The scalar-table SHA-256 values, with binary64 table encoding fixed by the accompanying source and manifest, are as follows. Line breaking of a printed digest has no mathematical significance.

```

Cosine  $q$ : a9992300d2bf71665aa2b6bd2727e798624cd297103bb200c7f0ca2baea55a2c
Cosine  $q''$ : 766b97d9ba81dff915acaf1c1ba9938166c3191011e744424c12fd3a141bd52
Two-mode  $q$ : f0be0477f6c4c98c7c3d879abf79f509fa637e9d4740fe8ee83bd4af52899580
Two-mode  $q''$ : aa35a51991d4a84c748e1908f7d95482ac507051d7c67b16c6919ee09f35377c

```

The cosine row has a completed ordinary execution of the distributed source. The two-mode row has completed ordinary and -00 executions: their entire receipts agree after removing only the optimization flag and the engine’s elapsed time. Source identities, table digests and machine-readable covering summaries accompany the package. These completed computations support (2.9) and (2.10) subject to the stated implementation and arithmetic trust boundary. An optimization-mode comparison is a consistency check, not an independent enclosure method.

## 6. A SEPARATE THIRTEEN-POINT OBJECTIVE

For the remainder of the supplement,  $k = k_{u_{13}}$  and  $q = k^2$  refer only to the six-mode profile in (2.3). Define

$$(6.1) \quad F(g) = V_{13, u_{13}}(g) + \sum_{j=1}^{12} p_j g_j, \quad g \in [0, \infty)^{12},$$

where

$$(6.2) \quad p_1 = p_{12} = p_e = \frac{811}{4800000}, \quad p_2 = \cdots = p_{11} = p_i = \frac{6359}{24000000}.$$

In particular,

$$(6.3) \quad P := \sum_j p_j = \frac{239}{80000}, \quad \varepsilon := \frac{31}{5000}.$$

We do *not* assert  $F \geq \varepsilon$  on this entire orthant. The first task is to restrict where a strict counterexample could lie. All scalar restrictions in the next two sections are necessary conditions under the explicit hypothesis  $F(g) < \varepsilon$ .

Write  $a = 1/(\sqrt{2} \sin z)$ . Since  $11/12 \leq \sin z/z \leq 1$ , we have  $1 \leq a \leq 12/11$ . The convenient full-density bounds

$$(6.4) \quad m := \frac{71}{100} \leq f_{u_{13}}(t) \leq M := \frac{311}{275} \quad (|t| \leq \tfrac{1}{2})$$

follow from  $\cos z \geq 3/4$  and  $\sum |u_j| = 4/125 \leq 1/25$ . The bounds apply to the perturbed density, not just its cosine part.

**6.1. Monotonicity and the minimum-gap bootstrap.** Differentiating the even Fourier integral gives

$$(6.5) \quad -k'(x) = 4\pi \int_0^{1/2} t f_{u_{13}}(t) \sin(2\pi xt) dt.$$

For  $0 < x \leq 1$  this is strictly positive. For  $1 \leq x \leq 27/25$ , the sine is at least  $1/2$  on  $[1/8, 3/8]$ : its argument belongs to  $[\pi/4, 81\pi/100] \subset [\pi/6, 5\pi/6]$ . This part of the integral in (6.5), before its factor  $4\pi$ , is at least  $71/3200$ . A negative contribution can occur only for  $t \geq 1/(2x) \geq 25/54$ ; its absolute value is at most  $M(1/2)(1/2 - 25/54) = 311/14850$ . Their difference is  $1183/950400 > 0$ . All other contributions are nonnegative. Hence  $k$  is strictly decreasing on  $(0, 27/25]$ .

The exact point evaluator in Section 3 verifies  $k(5/6) > 0$  and  $k(5/6)^2/6 > \varepsilon$ . If some gap were at most  $5/6$ , monotonicity would make its nearest term alone exceed the target. Thus all gaps of a strict counterexample exceed  $5/6$ . More generally, suppose all such gaps already exceed  $d_0$ , and  $d_0 < d_1 \leq 1$  with  $k(d_1) > 0$ . A gap at most  $d_1$  would imply

$$(6.6) \quad F(g) \geq \frac{k(d_1)^2}{6} + d_0 P.$$

The following outward lower endpoints exceed  $\varepsilon$  and prove the successive improved cuts. The pressure  $d_0 P$  includes all twelve gaps exactly once; only one nonnegative nearest energy term is kept.

| $d_0$    | $d_1$    | Lower endpoint in (6.6)    |
|----------|----------|----------------------------|
| 5/6      | 9/10     | 0.007009907197317624826482 |
| 9/10     | 91/100   | 0.006608495667780589234137 |
| 91/100   | 183/200  | 0.006356600267380480502615 |
| 183/200  | 917/1000 | 0.006262235820193320264541 |
| 917/1000 | 459/500  | 0.006214284540455943193710 |

Set  $\tau_0 = 209/200$ . Exact evaluation encloses

$$(6.7) \quad k(\tau_0) \in [0.015366680647566453432485, 0.015366680647566453432486]$$

and gives

$$(6.8) \quad \frac{k(\tau_0)^2}{6} - p_i(\tau_0 - 459/500) > 0.000005706104020715559434.$$

For  $459/500 \leq g \leq \tau_0$ , monotonicity, positivity, and (6.8) imply  $q(g)/6 + pg \geq p\tau_0$  for either pressure  $p$ . For  $g \geq \tau_0$  pressure alone proves the same assertion. Thus this first scalar baseline holds on the entire half-line  $[459/500, \infty)$ .

If all gaps exceed  $d_0$  but one is at most  $d_1 \leq \tau_0$ , use the eleven other scalar baselines to obtain

$$(6.9) \quad F(g) \geq \frac{k(d_1)^2}{6} + \tau_0 P - p_i(\tau_0 - d_0).$$

The steps  $(d_0, d_1) = (.918, .920), (.920, .922), (.922, .924)$  have respective outward lower endpoints

$$\begin{aligned} &0.006456625078131788200580, \\ &0.006352660139980702315412, \\ &0.006250599414089590385868. \end{aligned}$$

They exceed  $\varepsilon$ . Consequently every strict counterexample has

$$(6.10) \quad g_j > \delta := \frac{231}{250} \quad (1 \leq j \leq 12).$$

Each decimal just displayed is a terminating rational lower bound, not a claim about an unbounded numerical error.

## 6.2. Compact necessary domains. Put

$$(6.11) \quad B = \varepsilon - \tau_0 P = \frac{49249}{16000000}, \quad W_p(x) = \frac{q(x)}{6} + p(x - \tau_0).$$

The nearest cost  $w_p(x) = q(x)/6 + px$  is at least both  $p\tau_0$  and  $px$ . Since all higher-distance terms are nonnegative, a strict counterexample satisfies

$$(6.12) \quad \sum_j p_j(g_j - \tau_0)_+ < B, \quad W_{p_j}(g_j) < B.$$

The second inequality follows by using the other eleven baselines. In particular its coordinates and span satisfy

$$(6.13) \quad \begin{aligned} g_1, g_{12} &< \tau_0 + B/p_e = \frac{3124439}{162200}, \\ g_2, \dots, g_{11} &< \tau_0 + B/p_i = \frac{16103731}{1271800}, \\ \sum_j g_j &< 12\tau_0 + B/p_e = \frac{623616}{20275}. \end{aligned}$$

The final assertion uses  $\sum g_j \leq 12\tau_0 + \sum(g_j - \tau_0)_+$  and  $p_j \geq p_e$ . It does not assume the gaps lie close to one or two.

## 7. THE EXACT ONE-DIMENSIONAL GATE

**7.1. A point enclosure plus an analytic Lipschitz bound.** For  $x > 0$ , integration by parts in the Fourier integral yields

$$(7.1) \quad k(x) = \frac{f(1/2) \sin(\pi x) - \int_0^{1/2} f'(t) \sin(2\pi x t) dt}{\pi x}.$$

Here  $f = f_{u_{13}}$  on its support; the endpoint contribution is explicit, so no smooth extension by zero is assumed. The cosine part decreases on  $[0, 1/2]$ . Hence

$$\begin{aligned} |f(1/2)| + \int_0^{1/2} |f'(t)| dt &\leq a + \sum_j |u_j| + 2 \sum_j j |u_j| \\ &\leq C := \frac{27751}{22000}. \end{aligned}$$

For the perturbation this uses  $\int_0^{1/2} |2\pi j \sin(2\pi jt)| dt = 2j$ . It follows that  $|k(x)| \leq C/(\pi x)$ . Differentiating (7.1) gives

$$k'(x) = \frac{f(1/2) \cos(\pi x)}{x} - \frac{k(x)}{x} - \frac{2}{x} \int_0^{1/2} t f'(t) \cos(2\pi xt) dt,$$

so

$$(7.2) \quad |k'(x)| \leq \frac{C}{x} + \frac{C}{\pi x^2}.$$

The integral formula and  $f \leq M$  also give  $|k'(x)| \leq \pi M/2$ .

For a closed rational interval  $I = [l, r]$  with  $l > 0$ , let  $c = (l + r)/2$ ,  $h = (r - l)/2$ , and let  $[k_{\text{lo}}, k_{\text{hi}}]$  enclose  $k(c)$  by Section 3. For an outward interval  $[\pi_{\text{lo}}, \pi_{\text{hi}}]$ , define

$$(7.3) \quad K = \min \left( 1, \frac{C}{\pi_{\text{lo}} l} \right), \quad L = \min \left( \frac{\pi_{\text{hi}} M}{2}, \frac{C}{l} + \frac{C}{\pi_{\text{lo}} l^2} \right).$$

Then  $k(I) \subset [v, w] = [k_{\text{lo}} - Lh, k_{\text{hi}} + Lh]$ . Put

$$(7.4) \quad A_{\text{lo}} = \max(0, v, -w), \quad A_{\text{hi}} = \min(K, \max(|v|, |w|)).$$

These bound  $|k|$  below and above on  $I$ . Therefore

$$(7.5) \quad \frac{A_{\text{lo}}^2}{6} + p(l - \tau_0) \leq W_p(x) \leq \frac{A_{\text{hi}}^2}{6} + p(r - \tau_0) \quad (x \in I).$$

Every operation here is on rational endpoints with outward rounding. This is a whole-interval argument derived from the derivative bound, not midpoint sampling.

**7.2. Closed partition and retained components.** For each of the two pressures, start with the closed interval  $[\delta, \tau_0 + B/p]$ . If the lower bound in (7.5) is at least  $B$ , exclude the whole interval. Equality is sufficient because the sublevel under consideration is strict. If its upper bound is less than  $B$ , retain the interval. Otherwise bisect it unless its width is at most  $1/16384$ ; such unresolved boundary intervals are retained. Any emergency budget exhaustion must also retain pending intervals. Neither uncertainty nor termination may silently discard a possible counterexample.

The distributed gate checker completed both closed partitions. The endpoint gate has 50 excluded, 72 strictly passing, and 7 unresolved terminal intervals. The interior gate has 55, 74, and 10 respectively. No emergency budget was exhausted. Across both pressures, each run visited 534 cells and used 519 distinct exact midpoint values. A preliminary envelope-only test, using  $0 \leq q \leq K^2$ , can settle a cell without a point evaluation. Exact sorting, shared endpoints, and total interval lengths checked that the final leaves partitioned each initial domain. The ordinary and -00 executions agree in their mathematical output.

Merging retained adjacent leaves gives two closed necessary components for each pressure. Their exact endpoints are recorded here so that no floating approximation is used to define a domain:

$$(7.6) \quad \begin{aligned} S_e^{\text{gate}} &= \left[ \frac{393403101917}{425197568000}, \frac{538442949829}{425197568000} \right], \\ L_e^{\text{gate}} &= \left[ \frac{183665052303}{106299392000}, \frac{8164814247361}{425197568000} \right], \\ S_i^{\text{gate}} &= \left[ \frac{770962919881}{833486848000}, \frac{1054606088081}{833486848000} \right], \\ L_i^{\text{gate}} &= \left[ \frac{22558120767}{13023232000}, \frac{20694035057199}{1666973696000} \right]. \end{aligned}$$

The following slightly enlarged domains suffice for the local arguments:

$$(7.7) \quad S_e = [459/500, 317/250], \quad S_i = [459/500, 633/500], \quad L = [5/3, \infty).$$

The inclusions of (7.6) in the corresponding sets of (7.7) are exact rational comparisons. Thus a strict counterexample has an S or L label at every coordinate. This does not label every arbitrary nonnegative twelve-tuple: the strict-counterexample hypothesis is indispensable.

## 8. ANALYTIC SHORT AND LONG BASELINES, WITH RESERVES

8.1. **The short baseline** 53/50. We first prove, independently of any gate, that

$$(8.1) \quad \frac{q(x)}{6} + px \geq \tau_S p, \quad \tau_S = \frac{53}{50}, \quad x \geq 459/500,$$

for either pressure  $p$ . For  $459/500 \leq x \leq 1$ , the sine in (6.5) is nonnegative. Exact integration gives

$$\begin{aligned} -k'(x) &\geq m \frac{\sin(\pi x) - \pi x \cos(\pi x)}{\pi x^2} \\ &\geq m \frac{-\cos(\pi x)}{x} \geq m \cos(\pi/10) > \frac{19m}{20} > \frac{3}{5}. \end{aligned}$$

Here  $x \geq .9$ ,  $x \leq 1$ , and  $\cos(\pi/10) \geq 1 - \pi^2/200 > 19/20$  using  $\pi^2 < 10$ . For  $1 \leq x \leq T := 1063/1000$ , split the integral at  $b = 1/(2x)$ . On  $[0, b]$  the positive part is at least  $m/(4\pi x^2)$ . On  $[b, 1/2]$ , use  $|\sin(2\pi xt)| \leq 2\pi x(t - b)$  and  $t \leq 1/2$  to bound the absolute negative part above by  $M\pi(x - 1)^2/(8x)$ . Multiplying by  $4\pi$  gives

$$(8.2) \quad \begin{aligned} -k'(x) &\geq \frac{m}{x^2} - \frac{M\pi^2(x - 1)^2}{2x} \geq \frac{m}{T^2} - 5M(T - 1)^2 \\ &= \frac{3}{5} + \frac{366235595129}{62148295000000} > \frac{3}{5}. \end{aligned}$$

This proves  $-k' > 3/5$  throughout  $[459/500, 1063/1000]$ .

The exact point evaluation

$$(8.3) \quad k(53/50) \in [0.001990173438104262028271, 0.001990173438104262028272]$$

has lower endpoint greater than both  $a_S := 19/10000$  and  $5p_i$ . On  $[459/500, \tau_S]$  we therefore have  $k(x) \geq k(\tau_S) > 0$  and

$$\frac{d}{dx} \{q(x)/6 + px\} = k(x)k'(x)/3 + p \leq -k(\tau_S)/5 + p < 0.$$

The nearest cost is at least its value at  $\tau_S$ , which is at least  $p\tau_S$ . For  $x \geq \tau_S$ , pressure alone proves (8.1). No derivative assertion beyond the stated finite interval is used.

There is also a useful explicit reserve. For  $459/500 \leq s \leq \tau_S$  put  $d = \tau_S - s$ . Integrating (8.2) gives  $k(s) \geq a_S + (3/5)d$ . Consequently

$$(8.4) \quad r_{S,p}(s) := q(s)/6 + p(s - \tau_S) \geq \frac{a_S^2}{6} + (a_S/5 - p)d + \frac{3}{50}d^2.$$

All three coefficients are positive for both pressures.

**8.2. The long baseline 203/100.** Put  $A = \sqrt{2} \cot z = 2z \cot z$ . The same elementary sine and cosine estimates give  $3/2 \leq A \leq 24/11$ . Differentiating (3.4), with  $v = \pi x$ , yields

$$(8.5) \quad k'_0(x) = \frac{\pi}{(2v^2 - 1)^2} \left[ (2Av^3 + (4 - A)v) \cos v - (2(A - 1)v^2 + 1 + A) \sin v \right].$$

On  $5/3 \leq x \leq 41/20$  we have  $v > 5$ ,  $\cos v \geq 1/2$ ,  $\sin v \leq \pi/20$ , and  $4 - A > 0$ . The bracket is therefore at least  $Q(v) = Av^3 - (\pi/20)(2(A - 1)v^2 + 1 + A)$ . Using  $\pi^2 < 10$ ,  $A - 1 \leq 13/11$ ,  $1 + A \leq 35/11$ , and  $v > 5$  gives

$$(8.6) \quad \frac{\pi Q(v)}{4v^4} \geq \frac{15}{82} - \frac{1}{2} \left( \frac{13}{550} + \frac{35}{27500} \right) = \frac{15}{82} - \frac{137}{11000} > 0.$$

For the first term we used  $\pi A/(4v) = A/(4x) \geq 15/82$ . Since  $Q > 0$ , replacing  $(2v^2 - 1)^2$  by the larger  $4v^4$  in the positive lower bound is legitimate. Hence (8.6) is a lower bound for  $k'_0$ .

The perturbation has absolute coefficient sum  $4/125$ , and therefore

$$|(k - k_0)'(x)| \leq 2\pi \int_{-1/2}^{1/2} |t| \sum_j |u_j| dt = \frac{2\pi}{125} < \frac{44}{875}.$$

Combining this with (8.6) proves

$$(8.7) \quad k'(x) \geq \frac{15}{82} - \frac{137}{11000} - \frac{44}{875} = \frac{3}{25} + \frac{589}{3157000} > \frac{3}{25} \quad (5/3 \leq x \leq 41/20).$$

The exact point evaluation is

$$(8.8) \quad k(203/100) \in [-0.008907305837779524800319, -0.008907305837779524800318],$$

so  $k(\tau_L) < -1/125$  for  $\tau_L := 203/100$ . For  $5/3 \leq x \leq \tau_L$ , put  $d = \tau_L - x$  and integrate (8.7) to get  $k(x) \leq -1/125 - (3/25)d$ . Thus

$$(8.9) \quad r_{L,p}(x) := q(x)/6 + p(x - \tau_L) \geq \frac{1}{93750} + (1/3125 - p)d + \frac{3}{1250}d^2.$$

The linear coefficient is positive, since  $1/3125 - p_i = 1321/24000000 > 0$ . For  $x \geq \tau_L$ , nonnegativity of  $q$  and pressure suffice. We have proved the entire half-line baseline

$$(8.10) \quad q(x)/6 + px \geq \tau_L p \quad (x \geq 5/3)$$

for both pressures. Neither this proof nor the short proof uses derivative samples.

**8.3. Pattern budgets and safe use of reserves.** Fix any S/L label pattern permitted by the gates, and put  $t_j = \tau_S$  on S coordinates and  $t_j = \tau_L$  on L coordinates. Define

$$(8.11) \quad A_{\text{pat}} = \sum_j p_j t_j, \quad \Delta_{\text{pat}} = \varepsilon - A_{\text{pat}}.$$

For a strict counterexample, the two baselines and  $w_{p_j}(g_j) \geq p_j g_j$  imply

$$(8.12) \quad \sum_j p_j (g_j - t_j)_+ < \Delta_{\text{pat}}, \quad g_j < t_j + \frac{\Delta_{\text{pat}}}{p_j}, \quad \sum_j g_j < \sum_j t_j + \frac{\Delta_{\text{pat}}}{p_e}.$$

If there are  $a$  long interior and  $b$  long endpoint coordinates, these constants have the closed form

$$(8.13) \quad \begin{aligned} A_{a,b} &= \frac{53}{50}P + \frac{97}{100}(ap_i + bp_e), \\ \Delta_{a,b} &= \frac{12133}{4000000} - \frac{97}{100}(ap_i + bp_e), \\ \sum_j t_j &= 12\frac{53}{50} + \frac{97}{100}(a+b). \end{aligned}$$

All 33 budgets,  $0 \leq a \leq 10$ ,  $0 \leq b \leq 2$ , are positive: the largest baseline is the all-long baseline, whose deficit is  $1083/8000000 > 0$ . Thus these scalar budgets alone exclude no complete label pattern, and they do not make arrangements with the same counts equivalent for the longer-distance energy terms.

There is a useful optional closed span bound from the upper gate endpoints  $U_{eS}, U_{eL}, U_{iS}, U_{iL}$  in (7.6). Let

$$\begin{aligned} Z_e &= (2-b)(U_{eS} - \tau_S) + b(U_{eL} - \tau_L), \\ Z_i &= (10-a)(U_{iS} - \tau_S) + a(U_{iL} - \tau_L), \\ z_e &= \min(Z_e, \Delta/p_e), \quad z_i = \min(Z_i, \max(\Delta - p_e z_e, 0)/p_i). \end{aligned}$$

Then  $\sum g_j \leq \sum t_j + z_e + z_i$ . Indeed, relax the positive excesses  $v_j = (g_j - t_j)_+$  to  $0 \leq v_j \leq U_j - t_j$ ,  $\sum p_j v_j \leq \Delta$ . Since  $p_e < p_i$ , exchanging interior excess for available endpoint excess improves the span at no greater cost. Spend the budget on the two endpoint coordinates first, then the interiors; this gives exactly  $z_e + z_i$ . This bound is stated non-strictly. It does not authorize an equality rejection appropriate only to the strict caps in (8.12).

The polynomial reserves (8.4) and (8.9) bound the nearest terms including their pressures. On a box they may replace a weaker lower bound for that same nearest term. They must not be added to an energy estimate that has already spent it. If all coordinates except  $j$  have nonnegative residual reserves  $\sigma_\ell$ , a valid further necessary cap is  $g_j < t_j + (\Delta - \sum_{\ell \neq j} \sigma_\ell)/p_j$ . The reserve for coordinate  $j$  is deliberately absent from the sum.

## 9. ONE ADJACENT-LONG-GAP CREDIT

**9.1. A pressure choice dominated by both actual choices.** For  $x \geq 5/3$  define

$$(9.1) \quad \phi_L(x) = \begin{cases} p_i(x - \tau_L), & x \leq \tau_L, \\ p_e(x - \tau_L), & x \geq \tau_L, \end{cases} \quad r_L^*(x) = q(x)/6 + \phi_L(x).$$

This is the minimum of the two actual long residuals. Hence it is nonnegative by (8.10) and no greater than either  $r_{L,p}(x)$ . The function  $\phi_L$  is increasing and continuous, with its only slope change at  $\tau_L$ .

The adjacent-long credit is the whole-domain inequality

$$(9.2) \quad r_L^*(x) + r_L^*(y) + \frac{2}{11}q(x+y) \geq K_L := \frac{3}{100000} \quad (x, y \geq 5/3).$$

Here is its analytic compactification and finite certificate. Let

$$(9.3) \quad P_L(d) = \frac{1}{93750} + \frac{1321}{24000000}d + \frac{3}{1250}d^2.$$

It increases for  $d \geq 0$ . If  $x \leq 39/20$ , then  $d = \tau_L - x \geq 2/25$  and  $r_L^*(x) \geq P_L(2/25) = 3043/10^8 > K_L$ . If  $x \geq \tau_L + K_L/p_e = 179033/81100$ , its pressure residual alone pays  $K_L$ . The same applies to  $y$ . It remains to cover the closed square

$$(9.4) \quad [39/20, 179033/81100]^2.$$

**9.2. The complete scalar tables and finite square.** For the six-mode profile the distributed generator completed scalar tables using grid  $G = 500$ , 128-bit precision, and 15393 entries. The unused prefix below index 462 has nonnegative lower bound zero and unavailable curvature  $-\infty$ ; entries from 462 onward were computed on the entire closed cells using Section 3. The complete-table digests are

$q$ : 720514E4C09259C32629BBA740B5F7E3D61A3E85F6389AD7AE27BC61C0BB305C  
 $q''$ : A67323D95F9407B91ACF10BB517C60D8F4B2BB9CB73DA8E4AFC2523630EA0CB2.

The package includes the generated tables and their source-bound generation record. Its format specifies the exact profile, closed-cell convention, entry count, precision, unavailable prefix and byte order. The generator rebuilds the enclosures; matching a stored digest alone does not establish them. No twelve-dimensional covering claim is made by this scalar calculation.

The square (9.4) is covered outwards by cell indices 975, ..., 1103, giving 129 choices for each coordinate. Let  $Q_i$  be its exact represented binary64 lower bound for  $q$  and put  $R_i = \max(0, Q_i/6 + \phi_L(i/500))$ . The maximum with zero is valid because the true residual is nonnegative. For cells  $(i, j)$  the test is the exact rational comparison

$$(9.5) \quad R_i + R_j + \frac{2}{11} \min(Q_{i+j}, Q_{i+j+1}) \geq K_L.$$

Both sum cells are present; they cover the full interval for  $x + y$ . The expression is symmetric, so all  $129 \cdot 130/2 = 8385$  unordered pairs suffice. Each of the four triple invocations below recomputed all 8385 comparisons before using an LL replacement, both in ordinary and optimized Python. The smallest lower bound was

$$(9.6) \quad \frac{23653871141765599661283}{634106827533765836800000000} > \frac{3}{100000}.$$

Together with the closed tails, this proves (9.2) subject to the completed-table arithmetic boundary described above.

A use of (9.2) replaces its two nearest residuals and their connecting lag-two energy term by  $K_L$ . It does not add  $K_L$  while retaining those terms unchanged. In a longer string, such replacements can be made on disjoint adjacent pairs. For example, maximum matching credits  $w_j$  along a path can be selected by the elementary recurrence  $D_0 = D_1 = 0$ ,  $D_v = \max(D_{v-1}, D_{v-2} + w_{v-1})$ . Two neighboring LL replacements sharing a middle residual may not be added. Only this disjoint replacement principle is needed below.

## 10. FOUR TRIPLE INEQUALITIES AND THEIR FINITE CORES

For labels  $T_1, T_2, T_3 \in \{L, S\}$  define the local triple cost

$$(10.1) \quad C_{T_1 T_2 T_3}(x, y, z) = r_1(x) + r_2(y) + r_3(z) + \frac{2}{11} \{q(x+y) + q(y+z)\} + \frac{1}{5} q(x+y+z),$$

using  $r_L^*$  for every L and the actual pressure residual  $r_{S,p}$  for each S. The four completed assertions are

| Labels  | Domains by label    | Lower bound               |
|---------|---------------------|---------------------------|
| $LLL$   | $L = [5/3, \infty)$ | $\lambda = 9/100000$      |
| $SLL_i$ | $S_i, L, L$         | $\mu = 293723/2400000000$ |
| $LSL_i$ | $L, S_i, L$         | $\mu = 293723/2400000000$ |
| $SLL_e$ | $S_e, L, L$         | $\mu_e = 28447/480000000$ |

The short domains are those of (7.7). Full reversal gives the corresponding  $LLS$  assertions. Exchanging only the two long arguments of  $SLL$  is not a symmetry of (10.1); the two adjacent-sum slots would change.

Every residual and every remaining energy term in (10.1) is nonnegative on its domain. This permits a single variable's reserve to pay the whole target in a tail. For the first three rows, write  $\eta$  for their target. The long core is

$$(10.2) \quad [17/10, \tau_L + \eta/p_e].$$

Below its lower endpoint,  $d \geq 33/100$  and  $P_L(d) \geq P_L(33/100) > \max(\lambda, \mu)$  by exact rational comparison. Above its upper endpoint,  $\phi_L(x) \geq \eta$ . For an interior S, only the core

$$(10.3) \quad [47/50, 633/500]$$

remains: for  $s \leq 47/50$ , the positive polynomial in (8.4) at  $d = 3/25$  already exceeds  $\mu$ . The upper endpoint is the enlarged gate, not an extrapolated derivative range.

The inherited completed endpoint mixed cover uses the cores

$$(10.4) \quad S : [103/100, 317/250], \quad L : [189/100, 203/100 + 28447/81100].$$

The long lower tail is paid because  $P_L(14/100) = 78495/1200000000 > \mu_e$ . For the short lower tail,  $d \geq 3/100$  and the last two terms of (8.4), with  $p = p_e$ , already exceed  $\mu_e$ . The long upper endpoint is precisely  $\tau_L + \mu_e/p_e$ . All these cores and tails are closed; any overlap is harmless, and there is no omitted boundary between them.

## 11. EXACT ADAPTIVE AND ATOMIC TRIPLE VERIFICATION

**11.1. An integer lower bound for whole boxes.** The triple covers reuse the completed grid-500 tables. Round each finite rational core outwards to inclusive cell indices and take their Cartesian product. For an adaptive box  $[a_j, b_j]$ , use the full sum ranges in (4.4), now for one, two, or three coordinates. A range minimum supplies the lower  $q$  value over every required distance.

The interval and LL-replacement tests can be implemented with integer fixed-point arithmetic of scale  $S = 2^{80}$ . Set  $T_i = \lfloor SQ_i \rfloor$ , regarding the binary64 value  $Q_i$  as its exact rational value. For a single coordinate cell a safe residual integer is

$$(11.1) \quad \max(0, \lfloor T_i/6 \rfloor + \lfloor S\phi(i/500) \rfloor),$$

where  $\phi$  is its actual short pressure residual or the piecewise long pressure in (9.1). Precompute this residual bound for every coordinate cell. For a wider coordinate interval, the implemented bound takes the minimum of those residual bounds over all its cells; each individual cell bound is valid on its entire closed cell. One could instead take a range minimum of  $T_i$  and use the leftmost pressure value, but that generally gives a weaker lower bound and is not the recorded residual-table rule. Flooring a negative pressure contribution means rounding toward  $-\infty$ , not truncating toward zero. Each of the two lag-two contributions and the lag-three contribution is also multiplied by its exact coefficient and floored downwards. Accept only if the sum of these integers is at least  $\lceil S\eta \rceil$ .

If two adjacent labels are L, one alternative bound replaces their two residuals and connecting lag-two term by  $\lfloor SK_L \rfloor$ , retaining the other residual and unused energy slots. When both adjacent pairs are LL, take the maximum of the two alternatives, not their sum. The ordinary interval lower bound is another alternative. Every accepted lower bound therefore uses each of the six slots of (10.1) at most once.

Bisect the widest nonsingleton integer coordinate range as in Section 4. Reversal can remove only boxes whose entire reflected box is represented by the retained half: for *LLL* and *LSL* the first and third label domains agree, whereas no internal reversal reduction is made in *SLL*. A valid implementation retains diagonal or overlapping ranges for further examination instead of discarding them on the basis of a center ordering.

**11.2. A rigorous derivative from three exact point values.** The preceding tests leave some atomic cells unresolved. The following exact quadratic test closes them without requiring a finer scalar grid. Let  $h = 10^{-12}$ . By (2.7),  $q'$  is Lipschitz with constant  $4\pi^2$ . The centered difference is its average over  $[x - h, x + h]$ , so

$$(11.2) \quad \left| q'(x) - \frac{q(x+h) - q(x-h)}{2h} \right| \leq \frac{4\pi^2}{2h} \int_{-h}^h |v| dv = 2\pi^2 h.$$

Using the rational point evaluator for  $q(x-h)$  and  $q(x+h)$ , and  $2\pi_{\text{hi}}^2 h$  as the error allowance, gives an exact rational interval for  $q'(x)$ . Cancellation may widen this interval, but cannot invalidate it. All required centers and their two probes are above .9. The guarded below-one formula in Section 3 is therefore available when a short midpoint is below one; an  $x \geq 1$  routine is not silently applied outside its domain.

**11.3. The full atomic Hessian and exact LDL test.** The long pressure kink is at  $\tau_L = 2.03$ , exactly grid boundary 1015. Every closed atomic grid-500 cell lies on one side. Its appropriate affine branch agrees with the true pressure at every point of the closed cell, including a shared endpoint. The short pressure is already affine. Thus on an atomic triple cell, with coordinates  $g = (x, y, z)$ , the true cost has affine pressure and Hessian

$$(11.3) \quad \nabla^2 C(g) = \sum_{e=1}^6 c_e q''(a_e^T g) a_e a_e^T,$$

where the six pairs  $(a_e, c_e)$  are

$$(11.4) \quad \begin{aligned} &((1, 0, 0), 1/6), \quad ((0, 1, 0), 1/6), \quad ((0, 0, 1), 1/6), \\ &((1, 1, 0), 2/11), \quad ((0, 1, 1), 2/11), \quad ((1, 1, 1), 1/5). \end{aligned}$$

For an atomic tuple  $(i, j, k)$ , the lag-two curvature ranges include both indices  $i+j, i+j+1$  and  $j+k, j+k+1$ . The lag-three range includes  $i+j+k, i+j+k+1, i+j+k+2$ . Let  $\ell_e$  be the minimum of the corresponding complete curvature range, interpreted as an exact rational lower bound. Then

$$(11.5) \quad M = \sum_e c_e \ell_e a_e a_e^T \preceq \nabla^2 C(g)$$

throughout the atomic cell. Missing or nonfinite curvature disables this test; it is never changed to zero to obtain positive pivots.

Let  $g_0 = ((2i+1), (2j+1), (2k+1))/(2G)$  be the rational center. Compute the LDL factorization  $M = LDL^T$  with exact rational arithmetic. If any pivot is nonpositive, this criterion gives no acceptance. When every pivot is strictly positive, Taylor's integral remainder on the segment inside the box gives

$$(11.6) \quad \begin{aligned} C(g) &\geq C(g_0) + \nabla C(g_0)^T (g - g_0) + \frac{1}{2} (g - g_0)^T M (g - g_0) \\ &\geq C(g_0) - \frac{1}{2} \nabla C(g_0)^T M^{-1} \nabla C(g_0). \end{aligned}$$

The second line minimizes the lower quadratic over all of  $\mathbb{R}^3$ . Neither a stationary point of  $C$  nor the minimizing point of that quadratic needs to be inside the cell.

Enclose  $C(g_0)$  and its gradient using the exact point values and (11.2). These evaluations use the actual unclipped residual functions in (10.1), not the maximum with zero used in an independent interval lower bound. For a gradient interval vector  $G$ , interval forward substitution gives  $Y = L^{-1}G$ . If  $Y_j = [l_j, u_j]$ , put

$$(11.7) \quad U = \sum_{j=1}^3 \frac{\max(|l_j|, |u_j|)^2}{D_j}.$$

All  $D_j$  are positive, so  $U$  is an upper bound for the quadratic debit despite dependencies between interval components. The exact rational acceptance condition is

$$(11.8) \quad C(g_0)_{\text{lo}} - U/2 \geq \eta.$$

No LL matching credit is added to this bound: the six original slots have already been used to define  $C$ . Failed pivots, inadequate lower bounds, domain-guard failures, or interrupted point evaluation produce no acceptance.

The adaptive cover is complete only when both its pending stack and its unresolved terminal list are empty. If a budget interrupt occurs during atomic work, the current box is restored to pending work. Node identities and exact covered atomic volumes are checked in addition to, not instead of, the source's exhaustive traversal.

## 12. COMPLETED TRIPLE EVIDENCE AND THE EXACT ASSEMBLY

**12.1. What was actually executed.** The completed endpoint mixed computation needed no atomic Taylor test. It visited 77 nodes with 38 splits and 39 accepted leaves: 37 interval leaves and 2 LL-replacement leaves. Ordinary and optimized executions completed the full outward core in (10.4).

The other three rows were completed with the atomic rule above, both in ordinary Python and with -00. The recorded counts agree in both modes:

| Case                   | Nodes | Splits | Interval | LL | Taylor | Reversal |
|------------------------|-------|--------|----------|----|--------|----------|
| <i>LLL</i>             | 4117  | 2058   | 1059     | 9  | 868    | 123      |
| <i>SLL<sub>i</sub></i> | 2553  | 1276   | 1106     | 2  | 169    | 0        |
| <i>LSL<sub>i</sub></i> | 263   | 131    | 119      | 0  | 0      | 13       |

Every row has one initial root and zero pending or unresolved terminal boxes. The maximum depths are respectively 27, 27, and 22. The outward core atomic volumes are respectively 80621568, 45441792, and 45441792. They are volumes of adaptive leaves, not counts of point samples. In the symmetric rows, the proved and reversal-accounted volumes are respectively  $78952385 + 1669183$  for *LLL* and  $42186303 + 3255489$  for *LSL<sub>i</sub>*; the *SLL<sub>i</sub>* cover uses no reversal reduction.

These are completed executions of the distributed sources. All 1037 Taylor-accepted atomic cells were checked with rational matrix lower bounds, LDL factors, center and gradient intervals, and quadratic debits. The compact receipts retain exact canonical digests and counts of these generated objects, not the detailed matrices or point cache. The optional **-detailed** switch saves those objects in a fresh execution. The ordinary and optimized receipts agree after removal only of the optimization flag and the three declared timing fields. Checking the Taylor cells alone would still not prove an entire cover: the interval leaves, LL leaves, legitimate symmetries, roots, splits and zero pending volume remain indispensable.

**12.2. The 78-term allocation.** Partition the twelve gaps into the four consecutive triples

$$(12.1) \quad (g_1, g_2, g_3), \quad (g_4, g_5, g_6), \quad (g_7, g_8, g_9), \quad (g_{10}, g_{11}, g_{12}).$$

Let  $C_b^{\text{actual}}$  denote (10.1) with the actual pressure at every coordinate, rather than the dominated long choice  $r_L^*$ . With  $A_{\text{pat}}$  from (8.11), direct allocation of terms gives

$$(12.2) \quad F = A_{\text{pat}} + \sum_{b=1}^4 C_b^{\text{actual}} + R, \quad R \geq 0.$$

The allocation can be checked by the following inventory:

| Distance in gaps   | Original slots | Four triples | Remainder |
|--------------------|----------------|--------------|-----------|
| $h = 1$            | 12             | 12           | 0         |
| $h = 2$            | 11             | 8            | 3         |
| $h = 3$            | 10             | 4            | 6         |
| $4 \leq h \leq 12$ | 45             | 0            | 45        |
| Total              | 78             | 24           | 54        |

Each used slot has its unchanged original coefficient, namely  $1/6$ ,  $2/11$ , or  $1/5$ . Each pressure occurs once, through its baseline plus its residual. All 54 unused energy slots are nonnegative. By (9.1), each certified worst-pressure triple is no greater than  $C_b^{\text{actual}}$ .

If there is exactly one interior S, its baseline deficit is

$$(12.3) \quad \varepsilon - A_{\text{pat}} = \frac{1083}{8000000} + \frac{97}{100}p_i = \frac{941723}{2400000000} = 3\lambda + \mu.$$

The S is the middle coordinate of its block at positions 2, 5, 8, 11, using  $LSL_i$ . At positions 3, 4, 6, 7, 9, 10 it is an end coordinate, using  $SLL_i$  or its reversed  $LLS_i$  assertion. The other three blocks are  $LLL$ . Thus both interior placements are proved, and

$$(12.4) \quad F \geq \frac{31}{5000} \quad \text{on every retained interior-one-short region.}$$

No additional positive uniform margin is claimed here.

For either endpoint S, the mixed bound is  $\mu_e$ . Its deficit is

$$(12.5) \quad \varepsilon - A_{\text{pat}} = \frac{143647}{480000000} = 3 \left( \frac{1}{12500} \right) + \mu_e.$$

Since  $\lambda - 1/12500 = 1/100000$ , the three all-long blocks now improve this to

$$(12.6) \quad F \geq 0.00623 \quad \text{on both retained endpoint-one-short regions.}$$

If all twelve gaps are L, all four triples use  $\lambda$  and

$$(12.7) \quad F \geq \tau_L P + 4\lambda = \frac{51397}{8000000} = 0.006424625.$$

There are thirteen labeled regions here: the all-long region and twelve one-short regions, seven classes under full reversal. This count is not a percentage of the full orthant that has been proved.

### 13. WHY THE MIXED CREDIT WAS REBALANCED, AND WHAT REMAINS OPEN

The stronger standalone interior mixed credit  $\mu_{\text{old}} = 365723/2400000000$  is false. This can be demonstrated without an optimizer or a cover: evaluate the exact rational triple

$$(13.1) \quad (x, y, z) = (1.05423, 2.009397, 2.026457).$$

The point evaluator of Section 3 and the exact cost (10.1) give

$$(13.2) \quad C_{SLL_i}(x, y, z) \in [0.000132480370716898210350, 0.000132480370716898210351].$$

The upper endpoint is below  $\mu_{\text{old}}$  and above the new target  $\mu$ . Both long coordinates are below  $\tau_L$ , so their worst-pressure residuals equal the actual interior-pressure residuals. The failure is therefore not an artifact of incompatible pressure choices. It refutes that local credit only, not the full objective (6.1).

The successful correction has a precise budget interpretation. Increasing each of the three  $LLL$  credits in a one-short decomposition from .00008 to .00009 lowers the required mixed credit by .00003. That is exactly  $\mu_{\text{old}} - \mu$ . The smaller mixed credit is established on its whole domain by the completed computations, not inferred from the single point (13.1).

Combining the scalar gate with (12.4)–(12.7) now gives the precise remaining boundary:

$$(13.3) \quad F(g) < 31/5000 \implies \text{at least two coordinates have a retained short label.}$$

Whether such a strict counterexample exists is unresolved here. The multi-short regions have not been certified. No universal thirteen-point floor, no extra global zero-count decimal, and no worldwide record-priority assertion follows from this local completion. In particular, the two seven-point inputs to the main manuscript remain exactly (2.9) and (2.10).

#### 14. REPRODUCTION AND THE SCOPE OF THE PACKAGE

A self-contained reproduction has two logically distinct checks. The first evaluates the main manuscript's scalar count expressions from the two exact row configurations. Such an evaluation is conditional on the local rows; it is not a replay of them. The second runs a complete local verifier with the actual profile, pressure, floor, whole-cell tables, acceptance rules, and exhaustive coverage. The distributed package identifies these roles separately as `verify_counts.py` and `verify_local_rows.py`. Its README and manifest specify the commands actually supplied, their optional supplementary stages, and the execution status of each distributed source.

For the thirteen-point local results, the mathematical reproduction sequence is: the rational scalar bootstrap and closed gate; the short and long point checks together with the analytic derivative proofs; generation or full revalidation of the six-mode scalar tables; the LL square; the endpoint mixed cover; the three final triple covers with their atomic rule; and the exact assembly. The distributed `build_supplement_tables.py` constructs only the scalar tables. The entry point `verify_supplement.py` separates the cases `baselines`, `LL`, `LLL`, `SLL_i`, `LSL_i`, `SLL_e`, and `witness`; none of these is a full twelve-gap cover. The scalar-gate command and the completed receipt filenames are specified in the accompanying README. The program `assemble_supplement.py` binds the gate and all four source-generated triple receipts to the distributed sources and scalar data, then checks the exact regional allocation. It is a consistency-and-assembly checker, not an independent validator of arbitrary supplied proof logs. The reconstruction commands establish its local premises. Every stage is described above. A finite table can be distributed as an acceleration, but a verifier must know how to establish each of its whole-cell enclosures. Simply matching a stored digest establishes only that the same table has been read.

All acceptance guards must remain active under optimized Python; they must use explicit checked conditions rather than removable assertions. A bounded run may finish with an honest incomplete status. A modified program needs its own source-bound completed receipt before its output is described as a fresh replay.

This distinction is especially important here. The available completed records establish the two adopted seven-point rows and the specific thirteen-point local regions above, subject to the explicit computational trust boundary. They do not establish a complete thirteen-point target. The supplement supplies a linear, independently readable proof of what those records mean, while leaving that final unresolved question visible.

#### REFERENCES

- [1] Ainta, *zeta-simple-zeros*, MIT-licensed source code, commit 040c5e899e658aed7b56a2a87f501798fe10761d, <https://github.com/ainta/zeta-simple-zeros/tree/040c5e899e658aed7b56a2a87f501798fe10761d>. The original scalar-cell and convex-tangent engine and the cosine seven-point verification are attributed to this source; adapted portions retain its license.